Run report

2026-10-31T202105Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	1,600,000
dim	12	dataset	jets
seed	1,378,196,606	data_seed	42
hidden_units	128	n_layers	3
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	objective	wasserstein
lambda_gp	10.00	mmd_subsample	16,384
mmd_test	-9.799×10^{-5}	best_epoch	12
compile_cache_warm	True		
variables	**Mass and hard scale (IRC-safe kinematics):** m [GeV], $\ln \rho$ **Continuous angularities (IRC-safe jet shapes):** $\lambda^1_{0.5}$, w, λ^1_2 **Splitting and 2-prong substructure:** z_g, $\tau_{21}^{(\beta=1)}$ **Hadronization, multiplicity and fragmentation (IRC-unsafe):** M, n_{ch}, f_{ch}, p_T^D, q		
mmd_sigmas_detector	$3.107 \times (\frac{1}{2},\ \frac{1}{\sqrt{2}},\ 1,\ \sqrt{2},\ 2)$		
mmd_sigmas_particle	$3.027 \times (\frac{1}{2},\ \frac{1}{\sqrt{2}},\ 1,\ \sqrt{2},\ 2)$		

Timing

Phase	Seconds	Share (%)	Detail
Deconvolve			
data	2.349	3.8	cache hit – train
train	53.742	86.3	train
transfer	5.289		train
compile	11.019		train
epochs	36.095		train
select	1.155		train
particle_mmd	0.459	0.7	train
save	0.265	0.4	train
evaluate	5.464	8.8	train
total	62.280	100.0	
IBU baseline			
parse_config	0.001	0.0	ibu
data	1.922	13.8	cache hit – ibu
ibu	3.022	21.7	niter=10 – ibu
evaluate	8.954	64.4	12 variables – ibu
total	13.900	100.0	
OmniFold baseline			
parse_config	0.001	0.0	omnifold
data	1.929	0.1	cache hit – omnifold
omnifold	1,791.715	98.9	niter=3, epochs=50 – omnifold
init	3.705		DataLoaders and the two MLPs – omnifold
unfold	1,755.485		MultiFold.Unfold – omnifold
iter1_step1	423.518		detector-level reweighting – omnifold
iter2_step1	423.347		detector-level reweighting – omnifold
iter3_step1	363.762		detector-level reweighting – omnifold
iter1_step2	236.250		particle-level reweighting – omnifold
iter2_step2	149.573		particle-level reweighting – omnifold
iter3_step2	158.986		particle-level reweighting – omnifold
reweight	1.982		evaluating weights on z_target – omnifold
evaluate	18.156	1.0	12 variables – omnifold
total	1,811.801	100.0	
Common			
parse_config	0.002	0.0	plot
data	2.417	7.8	cache hit – plot
load	1.393	4.5	plot
plots	27.327	87.8	plot
total	31.139	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein Distance [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
m [GeV]	208.9	138.8	33.6	**5.038**	**97.6**	13.65	93.5
$\ln \rho$	227.9	20.71	90.9	**11.00**	**95.2**	19.11	91.6
$\lambda^1_{0.5}$	262.6	12.84	95.1	**8.404**	**96.8**	22.71	91.3
w	265.6	9.978	96.2	**8.218**	**96.9**	20.67	92.2
λ^1_2	255.4	9.215	96.4	**9.003**	**96.5**	13.82	94.6
${z_g}^\dagger$	60.45	60.45	0.0	19.71	67.4	**6.120**	**89.9**
$\tau_{21}^{(\beta=1)}$	121.8	70.10	42.4	26.84	78.0	**17.00**	**86.0**
M	224.2	177.3	20.9	**7.847**	**96.5**	7.850	96.5
n_{ch}	220.8	172.3	22.0	**8.967**	**95.9**	11.33	94.9
f_{ch}	52.86	40.94	22.6	**7.647**	**85.5**	9.369	82.3
p_T^D	256.8	30.80	88.0	**8.908**	**96.5**	12.05	95.3
q	79.67	17.51	78.0	9.307	88.3	**5.318**	**93.3**

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
m [GeV]	6.762	3.666	45.8	**0.06991**	**99.0**	0.1266	98.1
$\ln \rho$	8.383	0.4017	95.2	0.1784	97.9	**0.1552**	**98.1**
$\lambda^1_{0.5}$	9.359	0.4714	95.0	**0.1460**	**98.4**	0.1927	97.9
w	9.139	0.4527	95.0	**0.1434**	**98.4**	0.1721	98.1
λ^1_2	7.776	0.1728	97.8	0.1440	98.1	**0.1388**	**98.2**
${z_g}^\dagger$	0.6723	0.6723	0.0	0.1260	81.3	**0.09433**	**86.0**
$\tau_{21}^{(\beta=1)}$	3.372	1.934	42.7	0.2222	93.4	**0.1407**	**95.8**
M	9.640	7.633	20.8	0.1141	98.8	**0.1130**	**98.8**
n_{ch}	7.512	6.314	16.0	**0.1070**	**98.6**	0.121	98.4
f_{ch}	1.033	0.7960	22.9	**0.09556**	**90.7**	0.1037	90.0
p_T^D	8.223	1.311	84.1	**0.09761**	**98.8**	0.1196	98.5
q	1.576	0.2512	84.1	**0.08309**	**94.7**	0.08816	94.4

VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
m [GeV]	26.93	14.62	45.7	**0.2634**	**99.0**	0.4856	98.2
$\ln \rho$	33.31	1.576	95.3	0.6934	97.9	**0.6035**	**98.2**
$\lambda^1_{0.5}$	37.19	1.877	95.0	**0.5775**	**98.4**	0.7643	97.9
w	36.32	1.800	95.0	**0.5655**	**98.4**	0.6803	98.1
λ^1_2	30.90	0.6583	97.9	0.5379	98.3	**0.5168**	**98.3**
${z_g}^\dagger$	2.683	2.683	0.0	0.5040	81.2	**0.3773**	**85.9**
$\tau_{21}^{(\beta=1)}$	13.44	7.712	42.6	0.8844	93.4	**0.5583**	**95.8**
M	38.25	30.31	20.8	0.4253	98.9	**0.4203**	**98.9**
n_{ch}	29.81	25.07	15.9	**0.4016**	**98.7**	0.4567	98.5
f_{ch}	4.125	3.177	23.0	**0.3822**	**90.7**	0.4145	89.9
p_T^D	32.65	5.211	84.0	**0.3858**	**98.8**	0.4740	98.5
q	6.261	0.9940	84.1	**0.3223**	**94.9**	0.3442	94.5

Sliced Wasserstein, joint [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
All	152.4	467.3	−206.5	**11.61**	**92.4**	12.17	92.0

Mean 1D distance over random directions, in Data standard deviations. IBU: product of its per-observable weights.

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
m [GeV]	111.7	**94.08**	**15.8**	99.65	10.8	102.7	8.1
$\ln \rho$	253.1	32.57	87.1	**15.26**	**94.0**	53.22	79.0
$\lambda^1_{0.5}$	303.0	18.71	93.8	**9.833**	**96.8**	39.31	87.0
w	284.1	12.29	95.7	**8.657**	**97.0**	34.50	87.9
λ^1_2	253.8	11.13	95.6	**8.947**	**96.5**	25.43	90.0
z_g †	113.4	113.4	0.0	**34.80**	**69.3**	38.07	66.4
$\tau_{21}^{(\beta=1)}$	259.2	150.5	41.9	**25.61**	**90.1**	90.34	65.1
M	256.8	204.8	20.2	**36.84**	**85.7**	42.81	83.3
n_{ch}	234.6	188.6	19.6	**37.80**	**83.9**	42.39	81.9
f_{ch}	45.78	45.33	1.0	10.86	76.3	**10.55**	**76.9**
p_T^D	236.3	55.36	76.6	**18.75**	**92.1**	26.97	88.6
q	79.37	18.15	77.1	10.57	86.7	**6.930**	**91.3**

JS divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
m [GeV]	1.963	11.86	−504.3	**1.915**	**2.4**	2.449	−24.8
$\ln \rho$	9.878	1.731	82.5	**0.1987**	**98.0**	0.5908	94.0
$\lambda^1_{0.5}$	12.34	2.009	83.7	**0.1830**	**98.5**	0.4427	96.4
w	11.39	1.465	87.1	**0.1934**	**98.3**	0.3639	96.8
λ^1_2	8.824	0.8335	90.6	**0.1637**	**98.1**	0.2054	97.7
z_g †	2.041	2.041	0.0	**0.2576**	**87.4**	0.2873	85.9
$\tau_{21}^{(\beta=1)}$	10.50	6.577	37.4	**0.3376**	**96.8**	1.815	82.7
M	7.842	12.83	−63.7	**0.3223**	**95.9**	0.4834	93.8
n_{ch}	6.447	9.773	−51.6	**0.2861**	**95.6**	0.3955	93.9
f_{ch}	1.025	0.9435	7.9	**0.1187**	**88.4**	0.1698	83.4
p_T^D	8.342	4.029	51.7	**0.2983**	**96.4**	0.3674	95.6
q	2.027	0.9958	50.9	0.1088	94.6	**0.1019**	**95.0**

VLC divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
m [GeV]	7.809	45.93	−488.2	**7.622**	**2.4**	9.740	−24.7
$\ln \rho$	39.27	6.889	82.5	**0.7796**	**98.0**	2.312	94.1
$\lambda^1_{0.5}$	48.98	8.014	83.6	**0.7268**	**98.5**	1.757	96.4
w	45.23	5.837	87.1	**0.7577**	**98.3**	1.433	96.8
λ^1_2	35.08	3.315	90.5	**0.6369**	**98.2**	0.8006	97.7
z_g †	8.086	8.086	0.0	**1.029**	**87.3**	1.149	85.8
$\tau_{21}^{(\beta=1)}$	41.75	26.17	37.3	**1.343**	**96.8**	7.185	82.8
M	31.10	49.85	−60.3	**1.243**	**96.0**	1.892	93.9
n_{ch}	25.58	38.38	−50.1	**1.107**	**95.7**	1.545	94.0
f_{ch}	4.095	3.766	8.0	**0.4742**	**88.4**	0.6782	83.4
p_T^D	33.05	16.05	51.4	**1.181**	**96.4**	1.461	95.6
q	7.982	3.921	50.9	0.3865	95.2	**0.3807**	**95.2**

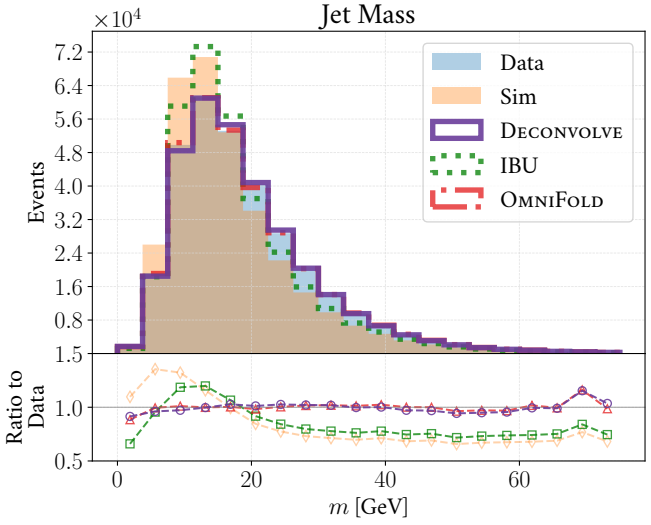
† IBU's purity binning produced a single bin, so IBU failed to unfold.

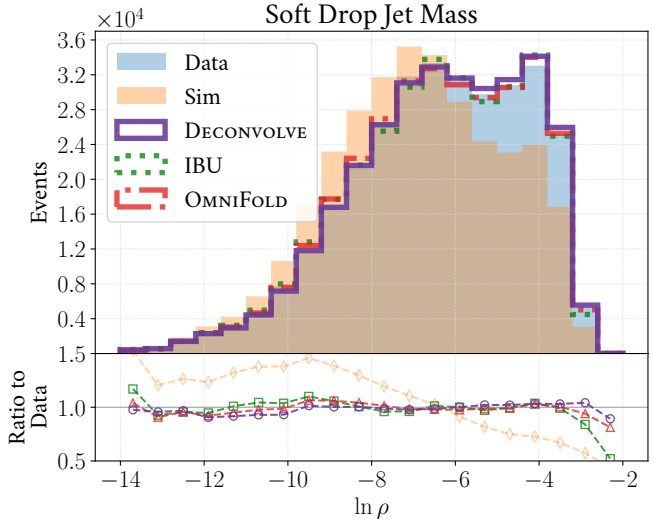
Sliced Wasserstein, joint [$\times 10^3$]

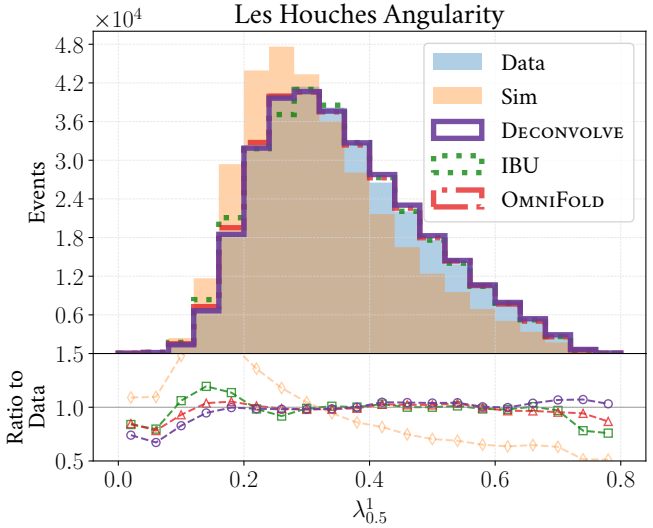
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
All	174.3	547.6	−214.2	**27.78**	**84.1**	41.44	76.2

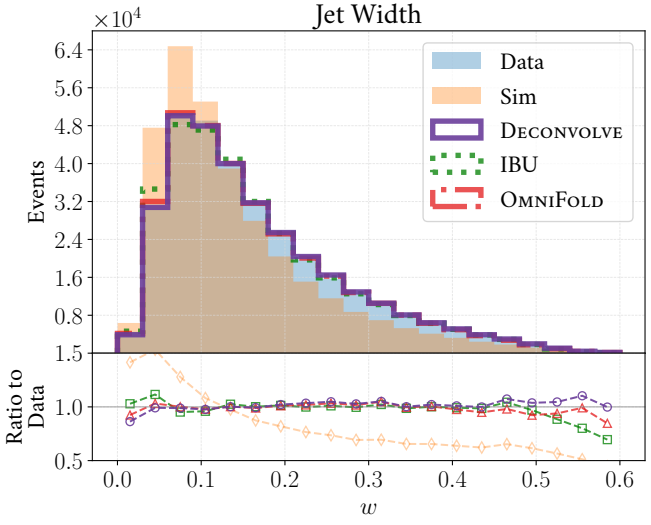
Mean 1D distance over random directions, in Truth standard deviations. IBU: product of its per-observable weights.

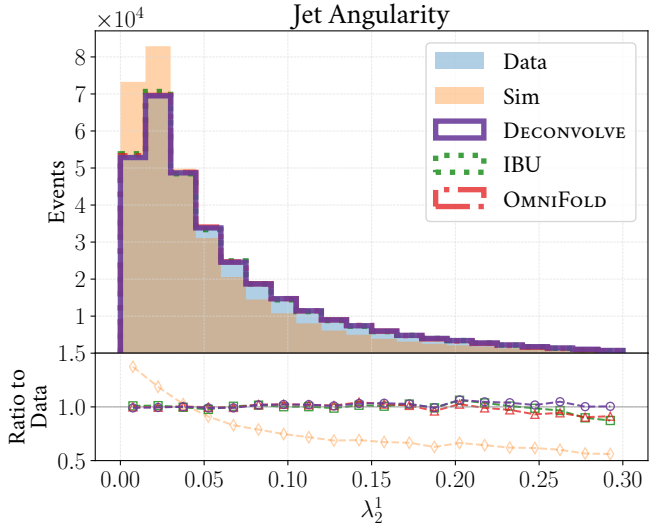
Detector Level (1 of 2)

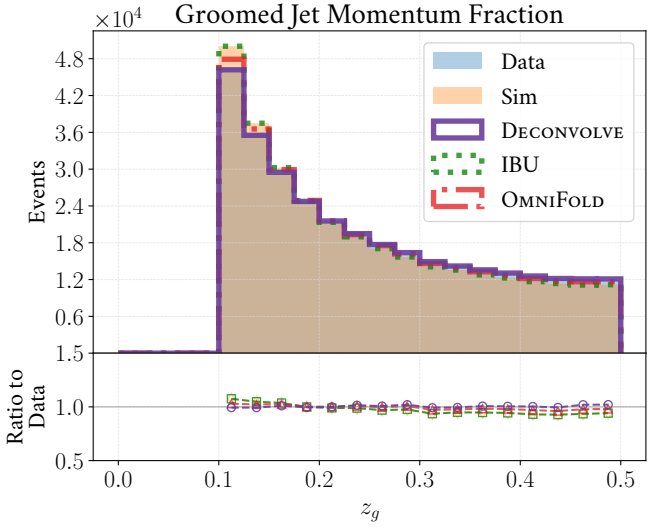




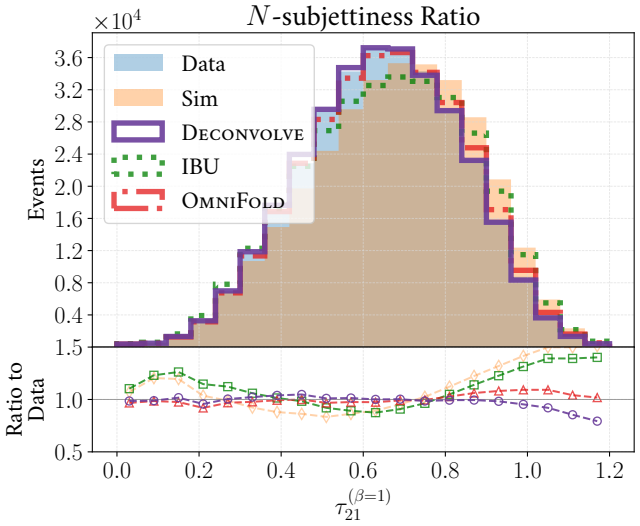


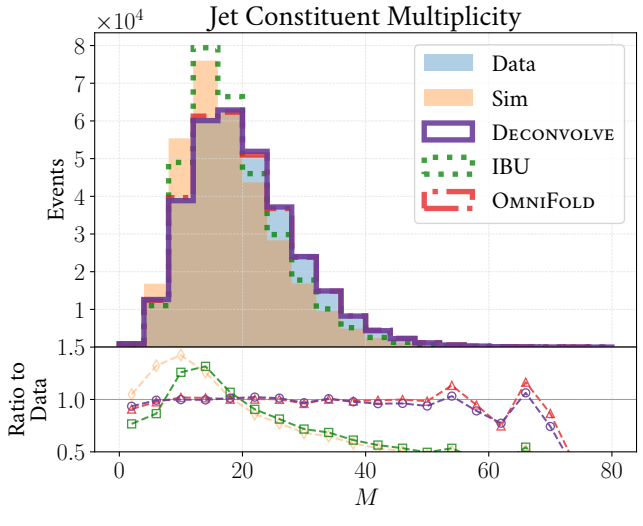


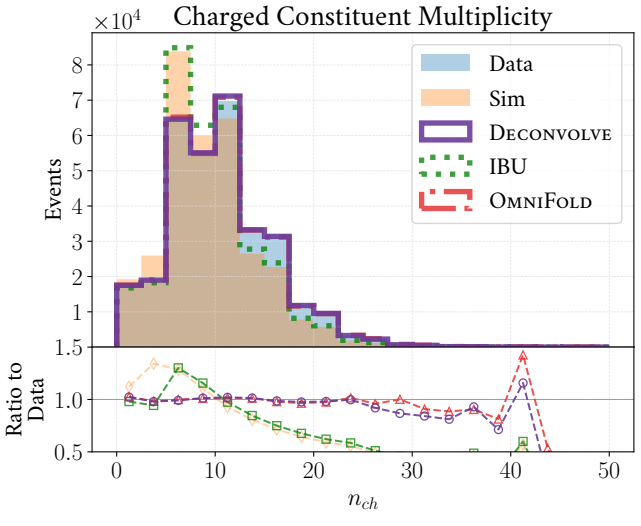


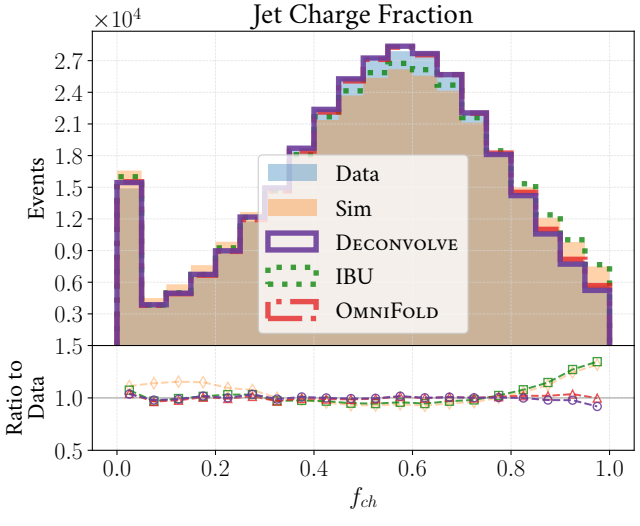


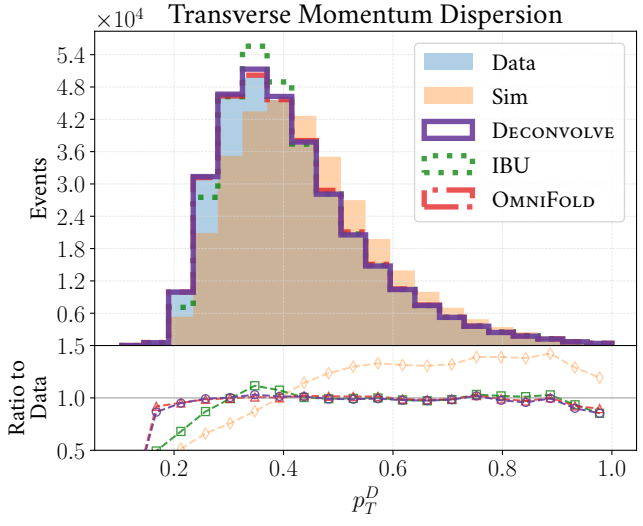
Detector Level (2 of 2)

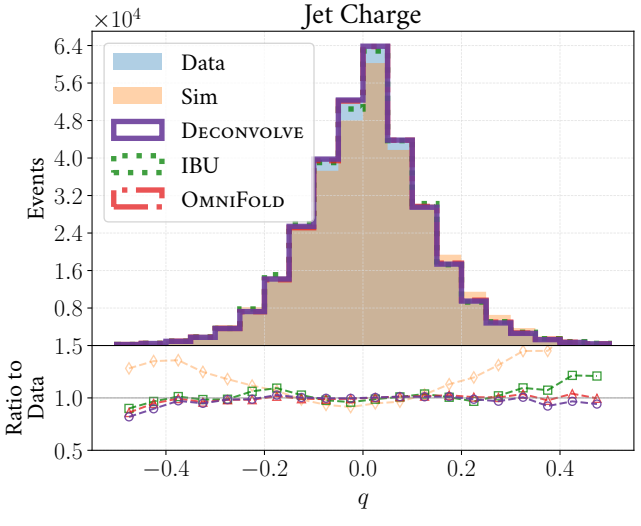




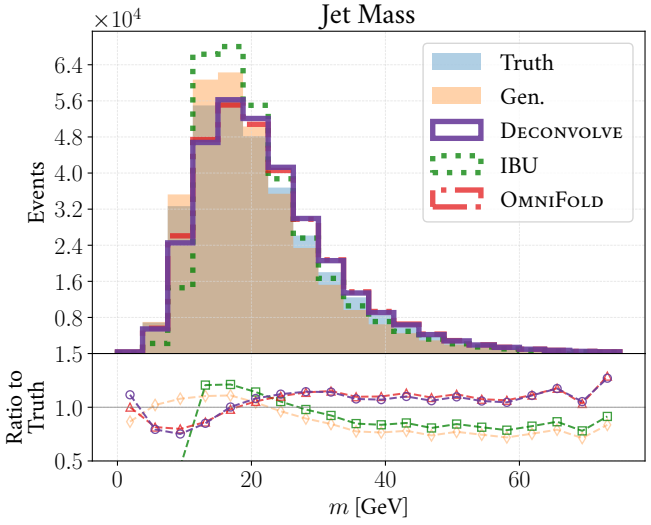


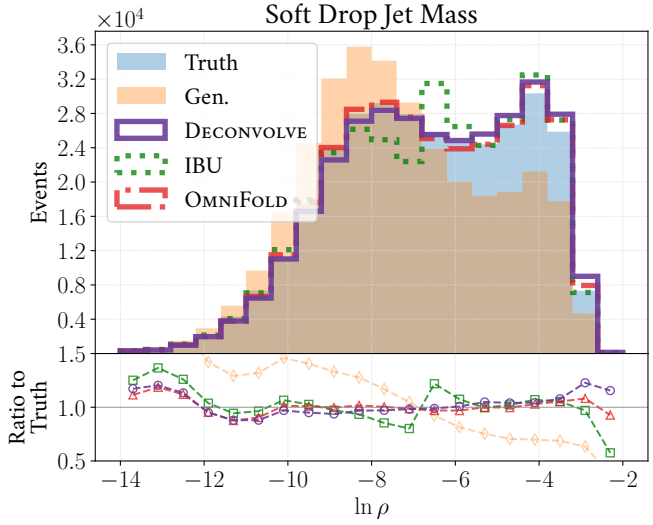


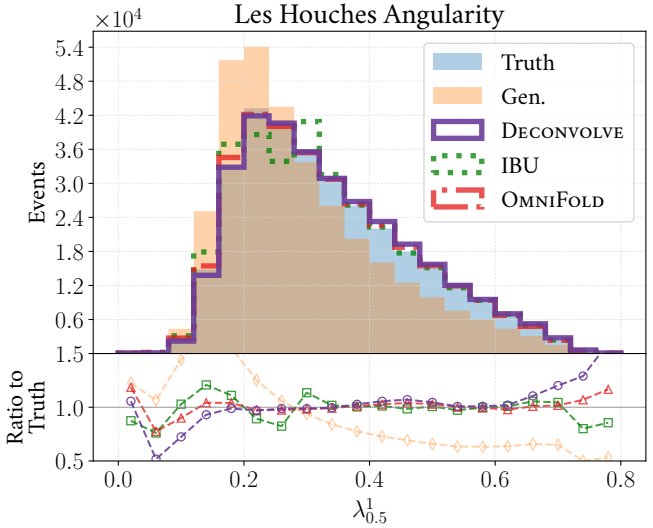


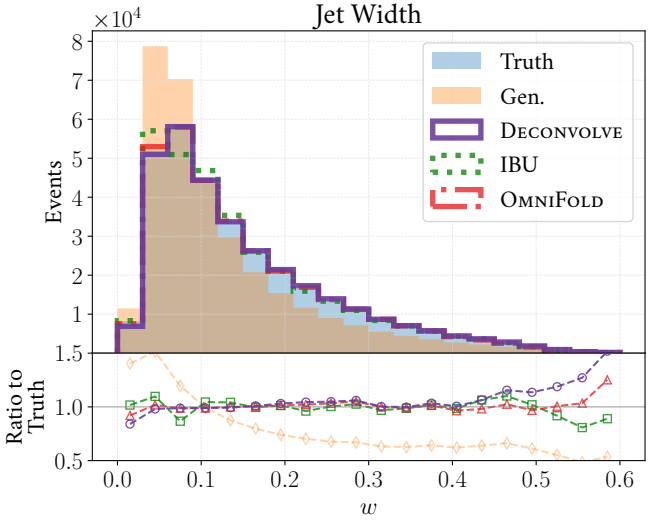


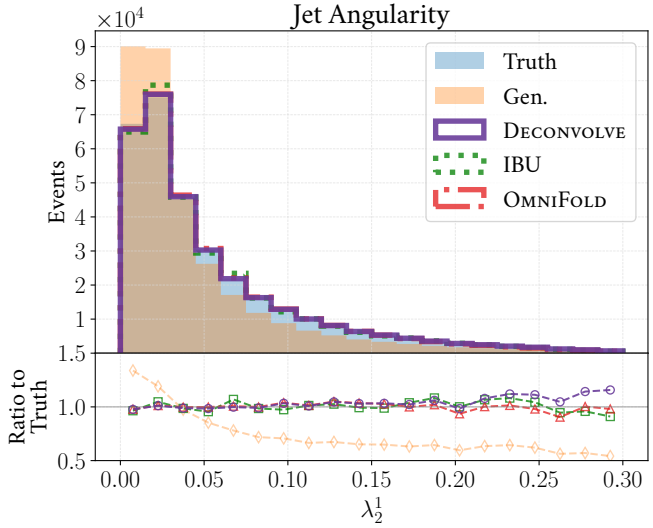
Particle Level (1 of 2)

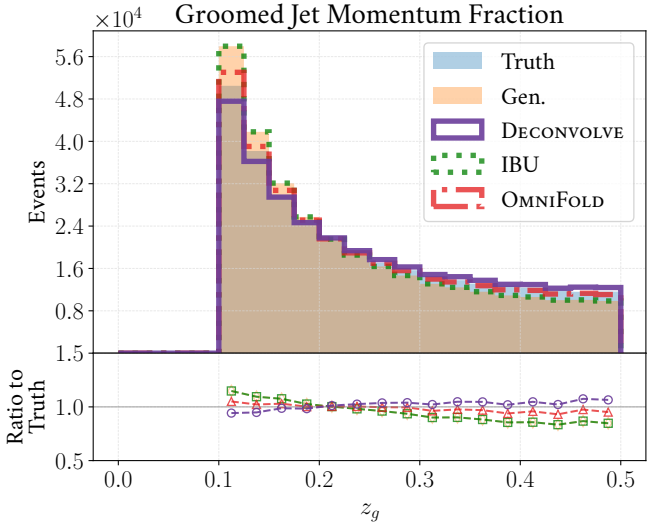




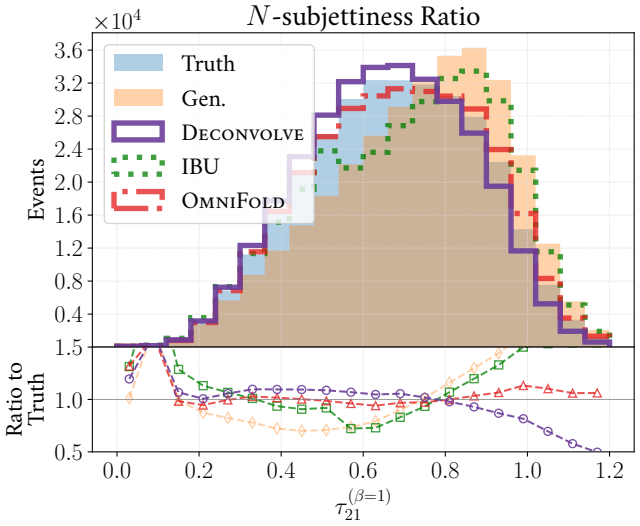


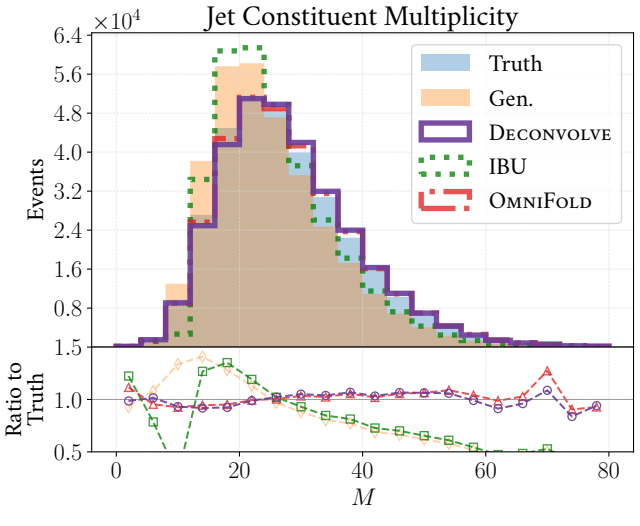


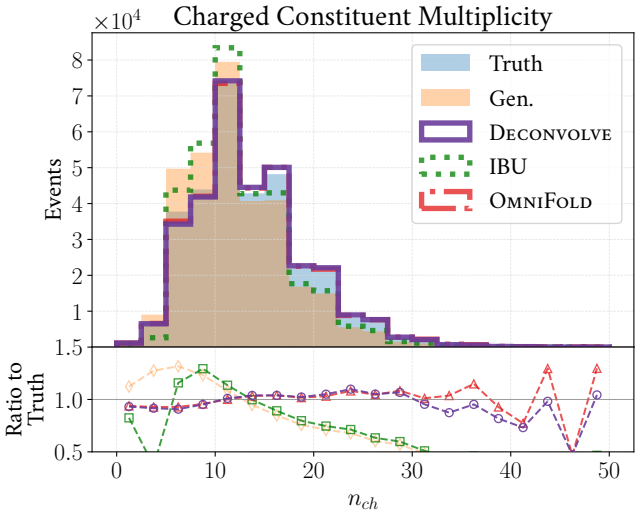


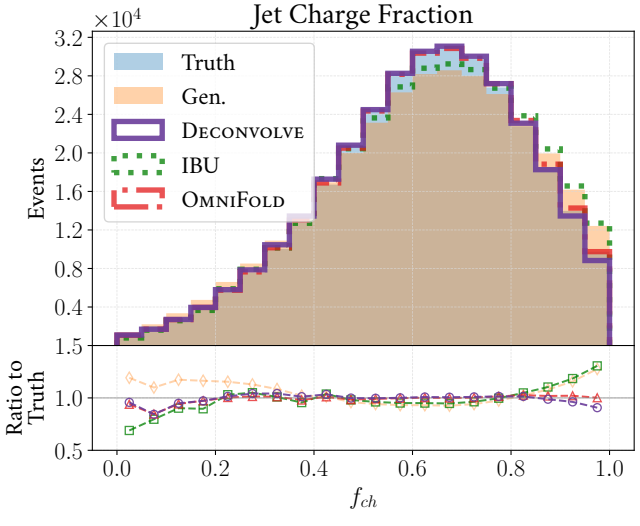


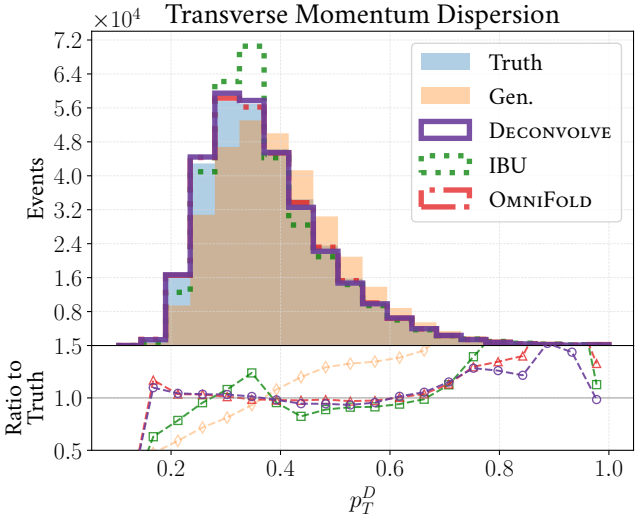
Particle Level (2 of 2)

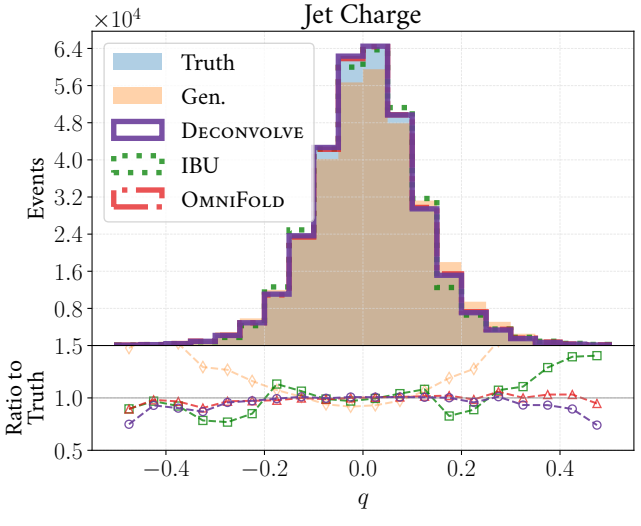












Training diagnostics

Training losses



Selection criteria

